

Module 12: Master Comparison: TCP vs UDP

Exhaustive 12-Point Comparative Matrix, Header Structural Divergence, Performance Metrics, and Practical Protocol Selection Guidelines.

Master Comparison: Transmission Control Protocol (TCP) vs User Datagram Protocol (UDP)

Comparison Parameter	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection Paradigm	Connection-Oriented (3-Way Handshake Setup)	Connectionless (Zero handshake setup)
Reliability	100% Reliable (Positive ACK, RTO Retransmissions)	Unreliable (Best-effort delivery; no recovery)
Data Framing Type	Byte-Stream (Continuous bytes, circular buffers)	Message-Oriented (Preserves packet boundaries)
Header Overhead	20 to 60 Bytes (Heavy overhead)	Strictly 8 Bytes Fixed (Extremely lightweight)
Packet Ordering	Guaranteed in-order arrival (Reordered via seq nos)	No ordering guarantee (Arrives out-of-order)
Flow & Congestion Control	Full Flow Control (rwnd) & Congestion (cwnd AIMD)	None (Transmits at application speed)
Transmission Paradigm	Strictly Unicast (One-to-One connection)	Unicast, Multicast & Broadcast Supported!
Speed & Latency	Slower due to handshakes, acks & window limits	Blazing fast; minimum possible latency
Real-World Applications	HTTP/HTTPS (80/443), FTP (21), SSH (22), SMTP	DNS (53), DHCP (67/68), VoIP, TFTP, Gaming

Architectural Decision Rule for Engineering Systems:

- Choose TCP when: Data accuracy and completeness are non-negotiable (Financial transactions, file download, web pages). Losing even 1 byte corrupts the file.
- Choose UDP when: Real-time timeliness matters more than 100% precision (Live video stream, voice call, FPS gaming). Retransmitting a dropped voice frame 500ms late is complete failure.
- Modern Hybrid (QUIC / HTTP/3): Runs on top of UDP to achieve zero handshake RTT, while implementing user-space reliability and stream multiplexing without Head-of-Line block.

1. 12-Point Architectural Divergence

Feature	TCP	UDP
Connection Setup	3-Way Handshake (1.5 RTT setup)	Connectionless (0 RTT instant dispatch)
Reliability	Guaranteed delivery with positive ACKs	Best-effort delivery without recovery
Framing	Continuous byte stream abstraction	Discrete datagram message boundaries
Header Overhead	20 to 60 Bytes	Fixed 8 Bytes

Transmission Scope	Strictly Point-to-Point Unicast	Unicast, Multicast, and Broadcast
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